FIG. 8A -- Gossip Replication Protocol (Page 1 of 2) 8/16

Replication is idempotent. Trust = ZK proof validity ONLY.
Quorum votes = durability acks, NOT validity votes. No BFT/PoW/PoS.

== SECTION A: GOSSIP MESSAGE ENVELOPE 800 ==

GossipMessage 800

messageId : string SHA-256(type:senderId:ts:extra)[0:32]
type : GossipMessageType (1 of 12 typed events)
senderId : string nodeId of ORIGINAL sender
timestamp : number ms since epoch (creation time)
payload : GossipPayload (union of 12 typed interfaces)
signature?: string HMAC-SHA256 hex (optional gossipSecret)

== SECTION B: TWELVE TYPED GOSSIP MESSAGES 802 ==

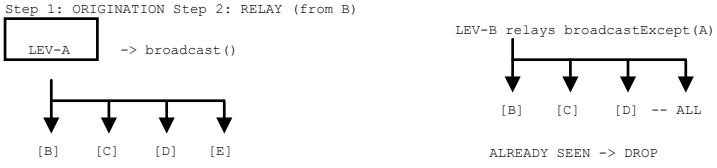
GROUP	TYPE	PAYLOAD SUMMARY	DELIVERY
PROOF LIFE-CYCLE (4)	proof:accepted	L2Receipt + AuditEvent + claimEntry? + roots + count	BROADCAST
	batch:built	L2Batch + L2Block + updatedReceipts/AuditEvents	BROADCAST
	batch:settled	batchNum + l1TxHash + roots	BROADCAST
	batch:failed	batchNum + reason	BROADCAST
DATA STORES (2)	message:stored	leafIndex + StoredMessage + MessagePayload + leafHash	BROADCAST
	state:snapshot	Full L2SequencerState (for IBD)	BROADCAST
LEADER -SHIP (2)	leader:heartbeat	nodeId + partitions[] + seqNums + pubKeyHex (5s/15s)	BROADCAST
	leader:promotion	newLeader + partIdx + prev + reason (timeout/manual/rejoin)	BROADCAST
QUORUM DURA-BILITY (3)	quorum:propose	proposalId + partIdx + seqNum + walEntryHash + stateDigest	BROADCAST
	quorum:vote	proposalId + voterId + walEntryHash + Ed25519 sig	DIRECTED (to leader)
	quorum:certificate	proposalId + votes[] (N/2 + 1) + stateDigest + leaderId	BROADCAST
DISC-OVERY (1)	peer:announce	nodeId + role + endpoint + partitions[] + capabilities	BROADCAST

CRITICAL: quorum:vote confirms DURABLE WAL WRITE, NOT proof validity.

== SECTION C: EPIDEMIC SPREAD MECHANISM 804 ==

5-node example: Leader A accepts proof

Step 1: ORIGINATION Step 2: RELAY (from B)



each marks messageId as seen After relay: ALL nodes have message.
Deduplication prevents infinite loops.

handleIncomingAndRelay(msg, from):

1. verifyGossip(msg) -- no secret? skip. HMAC mismatch? REJECT
2. seenMessageSet.has(msgId)? YES -> return false (duplicate)
3. markSeen(msgId) -- add to set + array
4. Notify handlers -- handler(msg) applies state mutations
5. If gossipRelayEnabled: broadcastExcept(msg, fromPeerId)